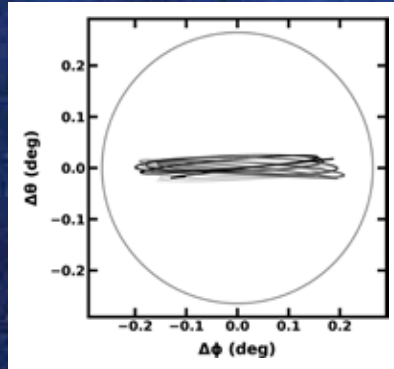


A dusty affair

Researchers working at the University of Utah have conceptualised an ingenious way of protecting the Earth from the wrath of the Sun's heat as more of it gets trapped in the atmosphere due to the over-accumulation of greenhouse gases. If their concept program is anything to go by, we could soon see dust being launched into space from the moon's surface or a space station orbiting our home planet. Yes, you read that right!

Their idea, which is based on the theory that is used to study planet formation around stars, suggests that we can take a small amount of material, in this case, dust, and launch it into a special orbit between the Sun and the Earth.



This would essentially cut down the amount of solar radiation reaching our planet, helping in the mitigation of the effects of climate change.

Commenting on the shield that the researchers are proposing, Sameer Khan, who is an undergrad

student at the university and a co-author of the study, said, "Because we know the positions and masses of the major celestial bodies in our solar system, we can simply use the laws of gravity to track the position of a simulated sunshield over time for several different orbits."

As absurd as it may sound, this theory has scientific evidence backing it. Yes, it is a very distant thought. However, if this is the ultimate solution to save the human race from the catastrophic effects of climate change, then after proper analysis of the side effects of this move, we could actually use it to protect our coming generations.

Link to Research Paper: <https://dgit.in/40UtzXt>

Origins: Meteorites

For the fans of the magazine, no, this is not an origins article about meteorites. A team of scientists and researchers from the Imperial College in London, after conducting a thorough analysis of meteorites, have found that some of Earth's volatile chemicals, some of which have played an essential role in the existence of life on our planet, have their origins in meteorites.

It may not come as a surprise to many, as in the past, meteorites have been known to be a source of multiple different elements occurring on our planet. However, if we tell you that zinc, which is one of the fundamental elements that we have around us, has its origins in space rocks, then your interest would peak, right? And what if we tell you that the asteroids that are believed to be the carriers of this element come from planets that we have a hard time seeing, like Jupiter, Saturn, and Uranus? Even more exciting, right? Well, this is exactly what the findings of the research study in focus suggest.

Commenting on the same, Professor Mark Rehkämper, said, "About half of Earth's zinc inventory was delivered by material from the outer Solar



System, beyond the orbit of Jupiter. Based on current models of early Solar System development, this was completely unexpected."

Research studies that were published earlier suggested that most of the materials contributing to the formation of the planet had their origins in the inner Solar System. These materials, in turn, were the sources of the volatile chemicals. However, these new findings suggest otherwise and could be our gateway towards understanding more about the planets far far away.

Link to the research paper: <https://dgit.in/meteor>